EduAirControl: Project Annexes

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Purpose

This folder gathers the supporting material behind the main deliverables: API contracts, architecture decisions, infrastructure files, risks, screenshots and the test inventory. They are unmodified copies of the project files, except for Annexes D and G, which were written for this delivery. Annexes D and G and this index exist in Spanish (`_ES`) and English (`_EN`).

Contents

Annex	Content	Location
A	OpenAPI 3 contracts of the services (8 files, including the shared component)	`Anexo_A_Contratos_OpenAPI/`
B	17 architecture decision records (ADR-001 to ADR-017)	`Anexo_B_Decisiones_Arquitectura/`
C	Docker Compose, wait and smoke-test scripts, Kubernetes manifests	`Anexo_C_Infraestructura/`
D	Project risk register (summary)	`Anexo_D_Registro_Riesgos_ES` and `Annex_D_Risk_Register_EN` (`.pdf` and `.md`)
E	39 screenshots used in the manuals	`Anexo_E_Capturas_Pantalla/`
F	Inventory of the 285 automated tests	`Anexo_F_Inventario_Pruebas/`
G	Glossary	`Anexo_G_Glosario_ES` and `Annex_G_Glossary_EN` (`.pdf` and `.md`)

Annex A. OpenAPI contracts

File	Service	Documented paths
`api-gateway.yaml`	API Gateway	2
`auth-service.yaml`	ms-security	12
`ms-classroom-management.yaml`	Environments and campuses	7
`ms-environment-monitoring.yaml`	Monitoring and analysis	10
`ms-sensor-management.yaml`	Sensors and devices	10
`ms-user-experience.yaml`	Favorites and preferences	8
`ms-user-management.yaml`	User profile	4
`_shared.yaml`	Shared components	–

Annex B. Architecture decisions

The ADR files themselves are written in English.

ADR	Topic
001	Documentation language

ADR	Topic
002	Bounded contexts and microservice boundaries
003	Database strategy
004	Communication patterns
005	API Gateway
006	Authentication and authorization (token flow partially superseded by 017)
007	Event delivery and consistency
008	Database migration tool (Liquibase)
009	Internal hexagonal architecture
010	Containerization strategy
011	Production deployment strategy
012	Observability stack
013	Circuit breaker adoption
014	Environmental sensor variables and thresholds
015	Folding alerts into ms-environment-monitoring
016	Multi-tenancy by institution and campus
017	OAuth2/OIDC authorization server and Backend-for-Frontend

Annex C. Infrastructure

- `docker-compose.yml`: orchestrates the services, PostgreSQL, Redis and RabbitMQ locally.
- `wait-for-services.sh` and `smoke.sh`: wait until the services are healthy and run an end-to-end check (register, login, logout and reuse of a revoked token).
- `k8s/`: namespace, configmap, ingress, PostgreSQL, RabbitMQ, Redis, applications and a secrets template (`secrets.example.yaml`).

Warning. These files contain sample development values only (for example `REPLACE_ME` or `change-me-in-vault`). No `.env` files or keys were copied. In a real environment, secrets must come from a vault and must not be repeated.

Annex E. Screenshots

The images `image.png` to `image-38.png` come from the technical, installation, administrator and user manuals. They keep their original names so the manuals can still link to them.

Annex F. Test inventory

`Inventario_Pruebas_Automatizadas.csv` lists every test method by service and class, its type and whether it needs Docker. The result column is `NOT RUN`: it is updated with the real execution (see the `07_Pruebas` folder).

Cited material that was not in the zip

The ESP32 firmware files (`esp32_sensor_sender.ino` and `infra/iot/esp32-node`, mentioned in the risk register) and the `iot/esp32/` sensor-test folder were not in the zip received. If they exist in the repository, they should be copied to an `Anexo_H_Firmware_IoT/` folder.